10/12 group meeting

The same Lagrangian Picture in Hyper Kähler and Hitchin Geometry.

§1 Overview: the same Lagrangian Picture

The main comparison of this talk.

cpt hol symp geom	Hitchin geometry
Beauville-Mukai	GL_n-Hitchin
Debarre	SL_n-Hitchin
dual Debarre	PGL_n-Hitchin

The common pattern is

Jacobians ⟶ kernels ⟶ dual quotients

More precisely, over the smooth locus:

$$\frac{\text{Beauville-Mukai}}{Jac(C_t)} \qquad \frac{GL_n\text{-Hitchin}}{Jac(C_a)}$$

both are relative Jacobian systems

For the kernel-type systems

$$\frac{\text{Debarre}}{\ker(alb: Jac(C_t) \to A)} \qquad \frac{SL_n\text{-Hitchin}}{\ker(Nm: Jac(C_a) \to Jac(\Sigma))}$$

both are relative Prym(-like) systems

For the quotient-type systems

$$\frac{\text{dual Debarre}}{Jac(C_t)/\widehat{A}} \qquad \frac{PGL_n\text{-Hitchin}}{Jac(C_a)/Jac(\Sigma)}$$

both are relative quotient abelian varieties

Ref J. Sawon, Lagrangian fibrations by Prym varieties (2020)

§2 Compact Lagrangian Fibrations : BM and Debarre

Def

X : $2n$-dim IHS
(i.e X : cpt Kähler, $\pi_1(X)=0$, $H^0(X,\Omega_X^2)=\mathbb{C}\sigma$)
σ : everywhere nondeg.

$\pi : X \to B$, $\dim B = n$ is Lagrangian fibration if $\dim F = n$, $\sigma|_F = 0$ for smooth fiber F.

Thm (Matsushita '99)
Every non-trivial fibration of an IHS is automatically Lagrangian fibration & general fiber is a cpx torus.

Example 1 Beauville-Mukai system.

S : K3, $C \subset S$: sm curve of genus g
$\Rightarrow |C| \simeq \mathbb{P}^g$ by R.R. $\mathcal{C}/\mathbb{P}^g$: universal family of curves.

$X = \overline{Pic}^d(\mathcal{C}/\mathbb{P}^g) \to |C| \cong \mathbb{P}^g$: BM system
($2g$-dim IHS of $K3^{[g]}$ type)

Fiber over $t \in \mathbb{P}^g = \overline{Pic}^d(C_t)$

Key Idea

Moving curves on a symp. surf + their Jacobian $\Longrightarrow$ Lagrangian fib on IHS.

Example 2 Debarre system.

A : Ab surf $C \subset S$: sm curve of genus $g+2$
$\Rightarrow |C| \simeq \mathbb{P}^g$ by R.R

$X = \overline{Pic}^d(\mathcal{C}/\mathbb{P}^g)$ is NOT IHS (not simp conn)

$\to$ Take the Albanese map / summation map
$Alb : X \to A \Rightarrow K := \ker(X \to A) \to \mathbb{P}^g$: Debarre
of $2g$-dim IHS
Kum_g-type

For sm fiber : $i : C_t \hookrightarrow A$

$$0 \longrightarrow K_t \longrightarrow J(C_t) \xrightarrow{alb} A \longrightarrow 0$$

$K_t = \ker(J(C_t) \longrightarrow A)$ Prym-like var.

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§3 Hitchin systems via Spectral curves

Def

Σ : smooth curve of genus $g \geq 2$

GL_n-Higgs bundle on Σ is a pair (E, Φ)

- $E \longrightarrow \Sigma$: holomorphic vector bundle of rank n
- $\Phi \in H^0(\Sigma, End(E) \otimes K_\Sigma)$: Higgs field
  i.e matrix valued hol 1-form.
  $\Phi_x : E_x \longrightarrow E_x \otimes K_{\Sigma,x}$

We say that (E, Φ) is stable if for all Φ-inv sub bundle $F \subseteq E$ we have $\frac{\deg F}{\text{rank } F} < \frac{\deg E}{\text{rank } E}$ (slope stab.) i.e $\Phi(F) \subset F \otimes K_\Sigma$

Fix rank n, degree d.

$\mathcal{M}_{Higgs} (= \mathcal{M}_{Dol})$:= the moduli space of stable Higgs bundles
Dolbeault

If $(n, d) = 1 \Rightarrow \mathcal{M}_{Higgs}$: quasi-proj var of dim $2n^2(g-1) + 2$

Key idea

$\mathcal{M}_{Higgs} \sim T^* Bun_{GL_n}$ Bun_{GL_n}: moduli of stable GL_n-bundles

☺ $T_E Bun \simeq H^1(\Sigma, End(E))$

$T_E^* Bun \simeq H^1(\Sigma, End(E))^\vee \underset{\text{Serre}}{\simeq} H^0(\Sigma, End(E) \otimes K_\Sigma)$

$\Rightarrow \mathcal{M}_{Higgs}$ is a hol symp mfd (non cpt?)

Spectral construction

non-abelian Higgs data (E,Φ) $\Longrightarrow$ abelian data on a spectral curve (C,L)

Abelianization?

Higgs filed $\Phi_x : E_x \longrightarrow E_x \otimes K_{\Sigma,x}$: $n\times n$-matrix valued 1-form.

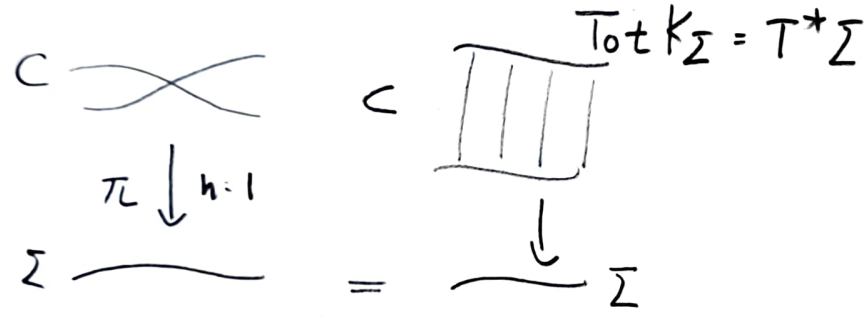
$\rightsquigarrow$ eigenvalues of this matrix will be n 1-forms.

fiberwise eigenvalues of Φ glued together

$\rightsquigarrow C \subset Tot(K_\Sigma) = T^*\Sigma$ spectral curve.

- def eq : $\det(\eta - \Phi) = 0$

$= \eta^n + a_1\eta^{n-1} + \cdots + a_n \quad a_i \in H^0(\Sigma, K_\Sigma^2)$



- each eigenvalue $\longrightarrow$ eigenspace (line)

$\rightsquigarrow$ glue together line bundle L over C

Prop Higgs bundle (E,Φ) can be recovered from its spectral data (C,L).

☺ $E = \pi_* L$, $\Phi : E = \pi_* L \longrightarrow \pi_*(L \otimes \underbrace{\pi^* K}_{\text{can. section of } \pi^*K \to Tot\, K_\Sigma})$

$= \pi_* L \otimes K = E \otimes K$ ☆

Def & Thm (Hitchin)

$h : \mathcal{M}_{Higgs} \longrightarrow \mathcal{A}_{GL} := \bigoplus_{i=1}^{n} H^0(\Sigma, K^i)$ Hitchin map

$(E,\Phi) \longmapsto (a_1, \cdots a_n)$ (or $(tr\Phi, tr(\Phi^2), \cdots tr(\Phi^n))$

is a holomorphic Lagrangian fibration

(algebraically completely integrable system.)

Fiber over $a \in \mathcal{A}_{GL}$ ($\rightsquigarrow C_a \subset T^*\Sigma$)

is $\mathrm{Pic}^{d'}(C_a)$ $\quad \boxed{\deg L = \deg E + n(n-1)(g-1)}$

Q BM v.s GL_n-Hitchin

BM	GL_n-Hitchin
S : K3	$T^*\Sigma$
$C_t \subset S$	$C_a \subset T^*\Sigma$
$\mathrm{Jac}(C_t)$	$\mathrm{Jac}(C_a)$
compact total sp.	non-compact total sp.

Thm (Donagi-Ein-Lazarsfeld '97)

the BM system degenerates to a compactification of GL_n-Hitchin system.

idea of pf We want $S \rightsquigarrow \overline{T^*\Sigma}$

$\Sigma \subset S$: sm curve (w/ $\mathcal{O}(\Sigma)$: very ample)

Note that $K_\Sigma = (K_S + \Sigma)|_\Sigma \simeq \mathcal{O}_\Sigma(\Sigma)$

$\Rightarrow N_{\Sigma/S} (\simeq \mathcal{O}_\Sigma(\Sigma)) \simeq K_\Sigma$ infinitesimal nbd.

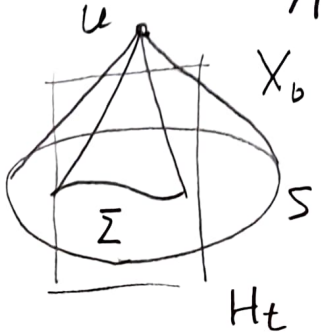
rough idea $S \rightsquigarrow \mathrm{Tot}(K_\Sigma) = T^*\Sigma$

deform to the normal cone

but $T^*\Sigma$ is non cpt

Then we consider $\overline{K_\Sigma} = \mathbb{P}(K_\Sigma \oplus \mathcal{O}_\Sigma)$ blow down ∞-section

(If Σ is non-hyper ellip) cone of $(\Sigma \underset{\text{can.}}{\subset} \mathbb{P}^{g-1})$



$X_b = \mathrm{Cone}(S) \subset \mathbb{P}^{g+1}$

$\Sigma = S \cap \mathbb{P}^{g-1}$: hyperplane section

- $H_t \not\ni v \Rightarrow X_0 \cap H_t \simeq S$.
- $H_0 \ni v \Rightarrow X_0 \cap H_0 = \mathrm{Cone}(\Sigma) = \overline{K_\Sigma}$

$$\begin{array}{ccc} \mathcal{X} & & t \neq 0 \quad\quad t = 0 \\ \downarrow & \text{s.t.} & S \quad\quad \overline{K_\Sigma} \\ \mathbb{P}^1 & & \cup \quad\quad \cup \\ \text{pencil} & & \Sigma \quad\quad \Sigma \end{array} \rightsquigarrow$$

(after blowup base locus Σ)

& curves in $|n\Sigma| \rightsquigarrow$ spectral curves in S_0.
(or additional curve). #

§4 From Jacobians to Prym's: the SL_n-Hitchin fibration

SL_n-condition

$(E, \bar{\Phi})$ { E: SL_n-bundle (i.e $\det E \simeq \mathcal{O}_\Sigma$) ← str gp.
$\bar{\Phi} \in H^0(\bar{\Sigma}, \mathrm{ad}(E) \otimes K_\Sigma)$
↑ adjoint bundle of E
(i.e $\mathrm{ad}(E) = \mathrm{End}_0(E) \Rightarrow \mathrm{tr}\bar{\Phi} = 0$)

Abelianization ⟹ $\pi : C_a \to \Sigma$

$\rightsquigarrow \mathrm{Nm}_\pi : \mathrm{Pic}(C_a) \to \mathrm{Pic}(\Sigma)$

$\mathcal{O}_{C_a}(\sum m_i p_i) \mapsto \mathcal{O}_\Sigma(\sum m_i \pi(p_i))$

$\det(\pi_* L) \simeq \mathrm{Nm}(L) \otimes \det(\pi_* \mathcal{O}_{C_a})$ ($\det(\pi_* L)$ fixed)

fixed $\Rightarrow$ fixed as $K^{n(n-2)/2} \to \det(\mathrm{Sym}(K_\Sigma^{-1}))$

Fiber $= \mathrm{Nm}^{-1}(K_\Sigma^{n(n-2)/2})$

$= \ker(\mathrm{Nm} : \mathrm{Jac}(C_a) \to \mathrm{Jac}(\Sigma))$

translate origin

Dehorre v.s SL_n-Hitchin

$C_t \hookrightarrow A$ $\qquad$ $C_a \xrightarrow{n:1} \Sigma$

$J(C_t) \xrightarrow{a|b} A$ $\qquad$ $J(C_a) \xrightarrow{Nm} J(\Sigma)$

DEL-type thm (de Cataldo-Mulik-Shen '22)

the Dehorre system ass to $|n\Sigma|$ w/ $g(\Sigma)=2$ degenerates to a compactification of the twisted SL_n-Hitchin system

§5 Duality and Mirror Symmetry in the Hitchin System

$$0 \to Prym(C/\Sigma) \to J(C) \xrightarrow{Nm} J(\Sigma) \to 0$$

dual $\rightsquigarrow$ $0 \to J(\Sigma) \xrightarrow{\pi^*} J(C) \to Prym(C/\Sigma)^\vee \to 0$

$$\Rightarrow \boxed{Prym(C/\Sigma)^\vee \simeq J(C)/\pi^* J(\Sigma)} \text{ quotient.}$$

PGL_n-Hitchin

Note $PGL_n = SL_n/\mu_n$ $(= GL_n/\mathbb{G}_m)$

w/ $\det(E\otimes F) \simeq \det(E)\otimes F^{\otimes n}$

$\hookrightarrow F \in J(\Sigma)[n]$

$$\Rightarrow \boxed{\mathcal{M}_{PGL_n} \simeq \mathcal{M}_{SL_n}/J(\Sigma)[n]} \text{ : orbifold}$$

Note that $PGL_n \longleftrightarrow SL_n$ Langlands dual

(conn. red semisimple) (root $\longleftrightarrow$ coroot)

	G_{sc}		G_{ad}
A_{n-1}	SL_n	$\longleftrightarrow$	PGL_n
B_n	$Spin_{2n+1}$	$\longleftrightarrow$	SO_{2n+1}
C_n	Sp_{2n}	$\longleftrightarrow$	PSp_{2n}

D_n $Spin_{2n} \longleftrightarrow PSO_{2n}$

$E_{6,7,8}$, F_4, G_2

Thm (Hausel - Thaddeus '03)

M_{SL} and M_{PGL} are (topological) mirror manifolds.
i.e. stringy Hodge numbers of M_{PGL} = Hodge numbers of M_{SL}.

MEMO

SYZ (Strominger - Yau - Zaslow) picture



fiber wise dual abelian var.

⇝ geometric Langlands.
Donagi - Pantev.

Via hyperkähler rotation
non-abelian Hodge theory $M_{Dol} \simeq_{\mathbb{R}\text{-an.}} M_{dR} \simeq_{\mathbb{C}\text{-an}} M_{Betti}$
(Riemann-Hilbert)
{Higgs} {flat conn} {repres}

$M_{dR} \longrightarrow A$ special Lagrangian fibration.

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§ 6 Back to the Compact Side: Dual Deligne Systems.

$$0 \to F_t = \ker(J(C_t) \to A) \to J(C_t) \to A \to 0$$

⇝ dual $\quad 0 \to \hat{A} \to J(C_t) \to F_t^\vee \to 0$

$$F_t^\vee \simeq J(C_t)/\hat{A} \simeq F_t/K(\lambda_{F_t})$$

Sawon Y-J. Kim ↑ kernel of pol.

dual Deligne : orbifold